1 Introduction

The purpose of the Lab 1 is to familiarize laboratory equipment, including the breadboard, power supply, multimeter, and Arduino board. We will firstly implement "voltage divider" experiment in order to get better understanding about breadboard and component connection which provides practical experience with the breadboard, multimeter, and power supply. Additionally, the lab will introduce the Arduino, where we will implement the pre-set "blink" program. This laboratory session aims to provide a fundamental understanding of the tools essential for diving into the field of mechatronics.

2 Experiment

In this laboratory session we will held two different experiments.
First experiment will provide better understand how we can connect components on the breadboard and analyzing the voltage divider principles.
Second experiment will be about Arduino setup, we will implement the "Blink" code which is pre-set in the Arduino.

2.1 Experiment 1

In our first experiment we will construct a basic voltage divider, that circuit will split the input voltage into a small output voltage according to the resistor values on the divider as shown in the Figure 1.

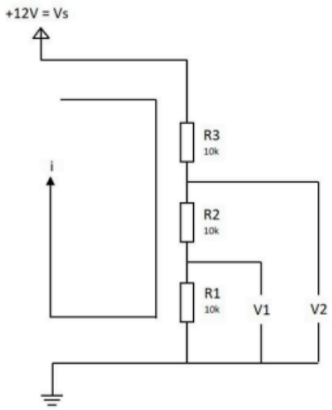


Figure 1: Voltage divider.

If we apply Kirchoff's Voltage Law on the circuit shown in Figure 1, we can obtain values of V1 and V2.

Kirchoff's Voltage Law provides us these equations.

$$V_s = (R_1 i + R_2 i + R_3 i) \tag{1}$$

$$V_s = (R_1 + R_2 + R_3) i \tag{2}$$

$$V_2 = V_s / (R_1 + R_2 + R_3) \tag{3}$$

$$V_2 = (R_1 + R_2) i \tag{4}$$

$$V_1 = (R_1) i \tag{5}$$

If we apply the Kirchoffs' Voltage Law to our circuit shown in the Figure 1, we must get with these equations.

$$12[V] = (10 + 10 + 10)[k\Omega] \times i[mA] -> i = 0.4[mA] \tag{6}$$

$$V2 = (20) \times 0.4 = 8[V] \tag{7}$$

$$V1 = (10) \times 0.4 = 4[V] \tag{8}$$

- First of all, we will construct the circuit shown in the Figure 2 below, with three 10kΩ resistors, one breadboard, some jumper cables, a multimeter to measure voltage values of V1 and V2 in order to check if the values are matching with the values we find on equations (7) and (8) while supplying +12V to the circuit.

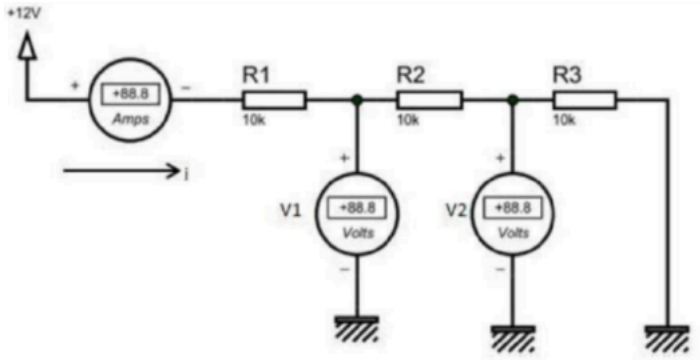


Figure 2: Voltage divider scheme.

- After constructing circuit shown in the Figure 2, we will that circuit on the breadboard as shown in the Figure 3.

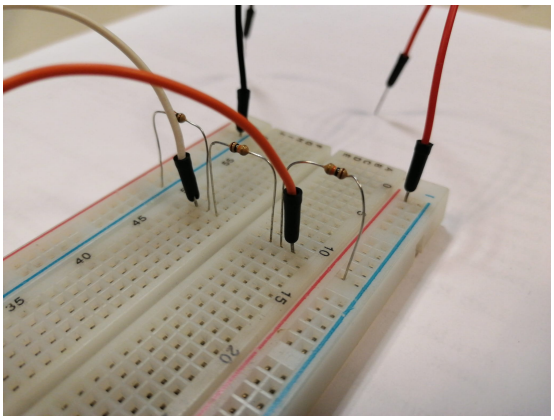


Figure 3: Constructed circuit on breadboard.

- Then we have measured the voltage difference between orange cable (V1) and ground as shown in the Figure 4.

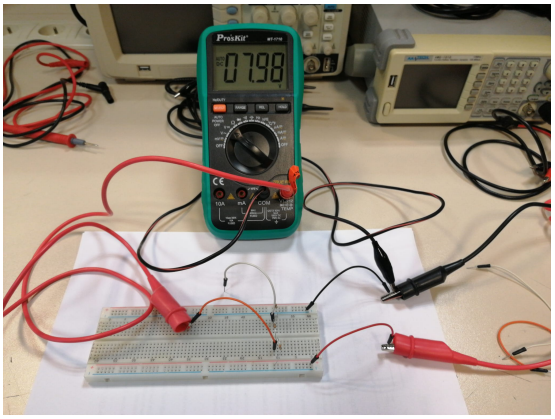


Figure 4: Measuring voltage difference on V1

Our measurements show that 0.798 Volts difference between V1 and ground so we can claim that

$$V1 = 0.798 \approx 8V$$

- Afterward, we measure the difference between white cable (V2) and ground as shown in the Figure 5.

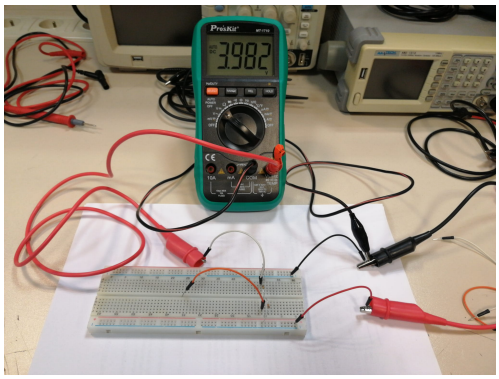


Figure 5: Measuring voltage difference on V2

Our measurements shows 3.982 Volts difference between V2 and ground so we can claim that.

$$V2 = 3.982 \approx 4V$$

- Lastly we measure the the current, we have decleared the current as 0.4mA previously. As

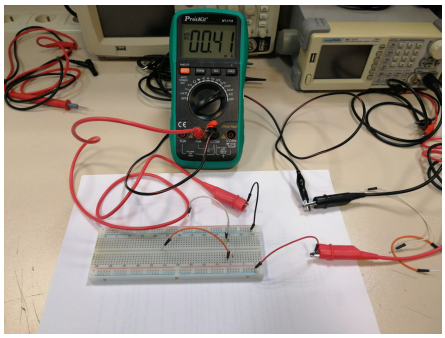


Figure 6: Measuring current on the circuit.

shown in the Figure 6, we measure the current as same as our calculations.

$$i = 0.41 \approx 0.4mA$$

After comparing our calculations and measurements, we can say we have built the voltage divider as it should be.

2.2 Experiment 2

Arduino is open-source electronic and it is easy-to-use and with that Arduino boards we can read inputs such as sensors or buttons.
In the second experiment we will use Arduino board and try to implement pre-set code "Blink" into the Arduino.

- First of all we downloaded Arduino IDE from the internet and completed the setup.
- Implemented pre-set code from: File → Examples → 01-Basic → Blink.
- Connected the Arduino board using a USB cable, then selected the board (Mega 2560) and the port from "Tools."
- Clicked on verify button to complete the code, then uploaded the code to Arduino.

  After implementation is done, small LED on the Arduino started to "Blink" as shown in the Figure 7. After changing the delay implemented in the code, we can observe the delay on the LED on the Arduino board.

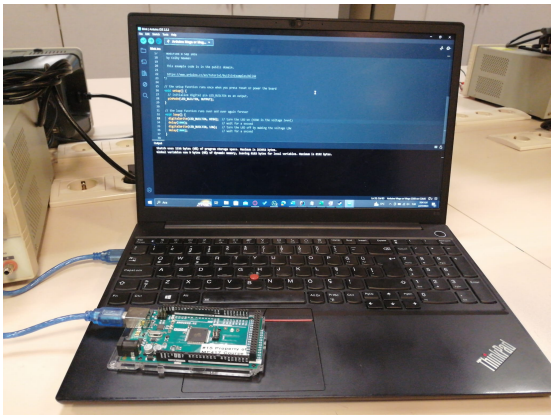


Figure 7: Arduino board with Blink code implemented.

3 Conclusion

Lab 1 served as introduction for exploration of mechatronics components, focusing on the usage of breadboard, power supply, multimeter, and Arduino board.

In the first experiment, we applied Kirchhoff's Voltage Law to understand and implement a voltage divider circuit. Constructing and measuring the circuit has improved our hands-on experiance on building circuits.

The second experiment provided Arduino's capabilities through the implementation of a basic "Blink" program. This hands-on exercise showed the board's capacity for processing inputs and generating outputs.

In summary, Lab 1 provided information about mechatronics, components and usage of components.